

Cambridge University Engineering Department

Engineering Tripos Part IIA

IIA Projects

PROJECTS: Interim and Final Report Coversheet

TO BE COMPLETED BY THE STUDENT(S)

Project:	SF4 Data Logger		
Title of report:	SF4 Interim Report Group Report		
Name(s): (capitals)	crsID(s):	College(s):	
OWEN MARLOVITS	om399	Corpus Christi	
RAHUL SHARMA	rs2288	Clare	
<u>Declaration</u> for: Interim Report 1 we confirm that, except where indicated, the work contained in this report is my/our own original work.			

Instructions to markers of Part IIA project reports:

Grading scheme

Grade	A*	A	B	C	D	E
Standard	Excellent	Very Good	Good	Acceptable	Minimum acceptable for Honours	Below Honours

Grade the reports by ticking the appropriate guideline assessment box below, and provide feedback against as many of the criteria as are applicable (or add your own). Feedback is particularly important for work graded C-E. Students should be aware that different projects and reports will require different characteristics.

Penalties for
lateness:

Interim Reports: 3 marks per weekday; Final Reports: 0 marks awarded – late reports
not accepted.

Guideline assessment (tick one box)

A*/A	A/B	B/C	C/ D	D/E

Markers:		Date:	
----------	--	-------	--

Delete (1) or (2) as appropriate (for marking in hard copy – different arrangements apply for feedback on Moodle):

(1) Feedback from the marker is provided on the report itself.

(2) Feedback from the marker is provided on second page of cover sheet.

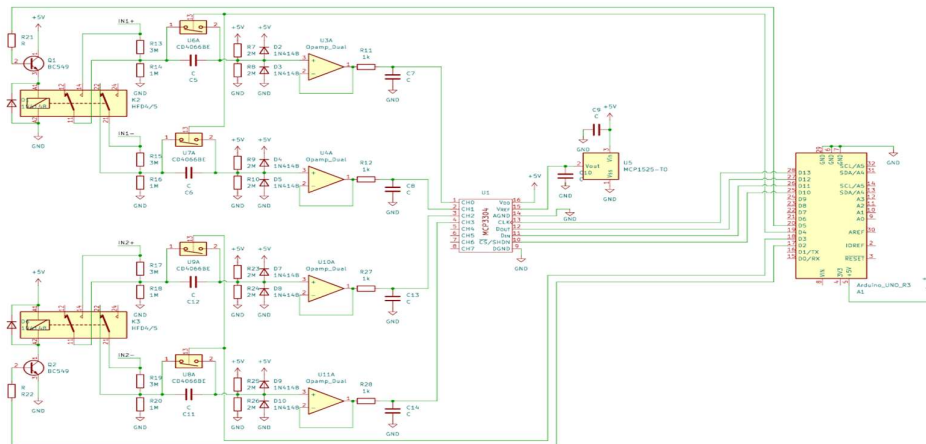
	Typical Criteria	Feedback comments
Project Skills, Initiative, Originality	Appreciation of problem, and development of ideas	
	Competence in planning and record-keeping	
	Practical skill, theoretical work, programming	
	Evidence of originality, innovation, wider reading (with full referencing), or additional research	
	Initiative, and level of supervision required	
Report	Overall planning and layout, within set page limit	
	Clarity of introductory overview and conclusions	
	Logical account of work, clarity in discussion of main issues	
	Technical understanding, competence and accuracy	
	Quality of language, readability, full referencing of papers and other sources	
	Clarity of figures, graphs and tables, with captions and full referencing in text	

We have decided to build a digital oscilloscope. This has scope for complicated analogue circuitry as well as a high level of digital signal processing.

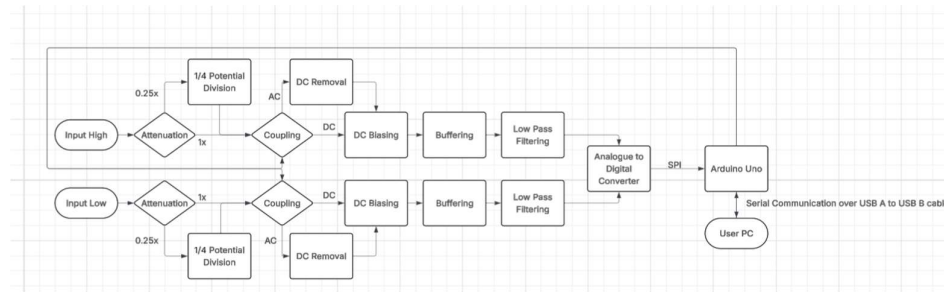
The core of the analogue circuit will be the Analogue-to-Digital Converter (ADC). For this, we have chosen the MCP3304, as it has four fully differential channels as well as a signed 13-bit output, theoretically giving a resolution of $\pm 0.61\text{mV}$. However, in order to stay within budget, we will only implement two differential channels. To ensure the readings are accurate, the MCP1525 is used as a voltage reference for the ADC. Each input will have an optional 4:1 attenuator, allowing voltages of $\pm 10\text{V}$ to be measured. There will also be an optional DC bias removal capacitor. Both of these can be controlled in software by the user. The input will be passed through an anti-aliasing filter and a buffer. Clamp diodes will be used to protect the input to the ADC. Due to a low sampling frequency and budget we have deemed oscilloscope probes unnecessary as crocodile clips should be sufficient for what we are doing.

The Arduino will communicate with the ADC via SPI and pass the information through to a computer through a serial link. The GPIO is also used to control the attenuator and DC bias removal for each channel. The bottleneck for the sampling rate is likely to be the serial communication, due to the speed of the Arduino and the communication protocol for serial transmissions. The computer will receive the data from the Arduino and plot it for the user to view. The interface will be graphical and have a number of features, including FFT, triggering, and RMS/peak-to-peak measurements.

Analogue Circuit Design

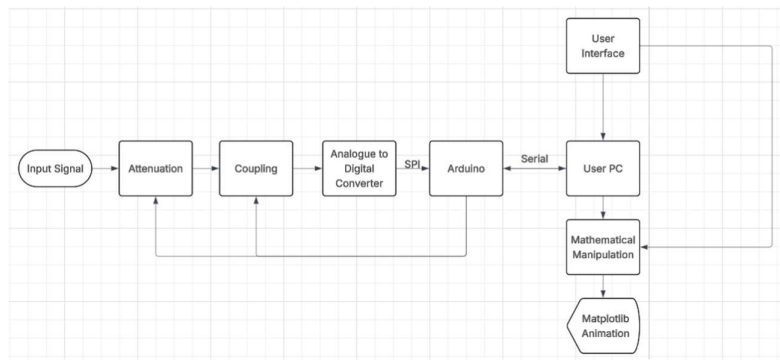


Hardware Block Diagram



The Block diagram above only focuses on one channel but our final setup will have two, in which case they are identical and come together at the Analogue to Digital Converter

Firmware/Software Block Diagram



Software Design/Features

- User Interface for controlling attenuation and coupling
- Continuous Time and Frequency representation of signals
- Automatic and Manual triggering
- Period, RMS, V_{pp} , V_{avg} measurements.
- Mathematical functions between channels ie multiplication and division.
- THD and SNR Calculations
- Total Power

ADC limited to 100KSps but serial link is a bottleneck so aiming for 20KSps

Parts List

Order Code	Part Name	Number Required	Cost per unit	Total Cost
1292235	MCP1525 Voltage Regulator	1	0.88	0.88
1605571	MCP6004-I/P Op Amp	1	0.46	0.46
1852026	MCP3304-BI/SL Analogue to Digital Converter	1	4.26	4.26
3879695	FIT0291 IC Adapter Board	1	0.70	0.70
4568806	CTM-30C Crocodile Clip	4	0.35	1.40
45555436	1N4148 Diode	10	0.04	0.44
3118236	CD4066BE Analogue Switch	1	0.72	0.72
4733577	HFD4/5 DPDT Relay	2	1.63	3.26
4710279	BC549 BBK NPN BJT	2	0.02	0.02
				Overall Cost: 12.16

We have not listed Passive components i.e. Resistors and Capacitors and this is only for the two channel configuration.

Division of Tasks

Rahul: Largely in charge of Analogue circuit and Firmware on the Arduino

Owen: In charge of PC receiver and Data visualisation and User Interface.

Testing

Ideally use a function generator, if unavailable use an oscillating pin on the Arduino and various combinations of R and C to filter given frequencies so that measurements may be taken by the oscilloscope to verify its behaviour. If this option is chosen another oscilloscope would be ideal to verify behaviour.